The Development of Attention: Research and Theory
James T. Enns (Editor)

Chapter 9

Texture Segregation in Young Children

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Developmental trends in texture segregation were explored in an effort to clarify a claim made in the adult literature (Callaghan, 1989) that this task invokes an attentional mechanism, and thus calls for a reconsideration of traditional preattentive/attentive dichotomy views of perceptual processing. In six experiments the degree of interference from irrelevant feature information and dimensional dominance in two-boundary ambiguous displays was assessed for the dimensions of hue, geometric form, and line orientation. Like adults, children show interference, but to a greater degree for hue and form stimuli, and little evidence of dimensional dominance. The implications of these results for revised views (e.g., Treisman & Gormican, 1988) of perceptual processing are considered.

In the perception of objects and events in the real world the process of seeing seems to be instantaneous, and not the product of a series of interconnected stages. The psychological study of object perception paints a different picture. A number of researchers (e.g., Kahneman, 1973; Neisser, 1967; Treisman & Gelade, 1980) have proposed models that incorporate the view that perception involves a series of processing stages. For these particular theorists, the most salient distinction between the stages involves the process of attention.

Kahneman (1973) proposes that a stage of unit formation, involving partitioning of the stimulus field according to Gestalt principles, precedes a stage where these units are recognized or attended to. Neisser (1967) coined the term preattentive to refer to this early stage of perception where processing was automatic, in contrast to the effort required of focused attention in later stages. Treisman (1982; Treisman & Gelade, 1980) extended this distinction in her feature integration theory of object perception. In this view, parallel and independent registration of features from across the visual field occurs in a preattentive stage, and is followed by an attentive stage where features are combined into identifiable objects through the process of focused attention on a specific spatial location.

The present research was inspired by the view that perception can be classified into two major stages, one that requires attention and thus effort (attentive stage), and one that does not (preattentive stage). In particular, the research is an exploration of the preattentive stage of perception using a texture segregation task and is designed to test the limits of the stage metaphor. This chapter begins with a brief review of the work that supports the preattentive/attentive stage distinction, followed by a look at recent evidence that challenges the dichotomy. This is followed by a consideration of how trends in developmental attention fit into a revised view of perceptual processing. The chapter ends with the presentation of developmental data from texture segregation studies and a discussion of the implications of the data for models of perceptual processing.

Support for the preattentive/attentive distinction

Treisman and her collegues (1982; Treisman & Gelade, 1980; Treisman & Paterson, 1984; Treisman & Schmidt, 1982; Treisman & Souther, 1985; see also Treisman, 1986 for a review of the work) have repeatedly provided empirical support for the preattentive/attentive distinction originally developed in the feature integration theory using a variety of converging operations. Treisman and Gelade (1980) reported that features (shape and colour) can be searched for in parallel across the visual field in search tasks, serve to promote easy detection of boundaries in texture segregation tasks, and can be erroneously combined to form illusory conjunctions in identification tasks. Treisman (1982) showed that providing observers with spatially homogeneous groups of items improves visual search for conjunctions, but not for features. Treisman and Schmidt (1982) used both verbal recall and simultaneous and successive matching tasks to show that unattended features (colour vs. shape and size vs. solidity) can be wrongly combined in illusory conjunctions. Treisman and Paterson (1984) reported that component lines and angles can form illusory arrows but not illusory triangles in a detection task and that triangles but not arrows promote easy texture segregation and pop out in visual search tasks. Finally, Treisman and Souther (1985) used a variety of stimuli to show that visual search for the presence of a distinguishing feature (e.g., circle with intersecting line target in regular circle background) is parallel, and is serial for the absence of the feature (regular circle in background of circles with intersecting lines). This form of visual search asymmetry is the latest method of convergence on the original view that single features effortlessly pop out in parallel from across the visual field, whereas multiple features or combinations of features require focused attention to stand out. The asymmetry suggests that the visual system codes deviations from a standard dimensional value, therefore, deviations themselves (e.g., circle with intersecting line) will pop out whereas the standard values themselves (e.g., circle) will not. Treisman and Gormican (1989) have recently undertaken a revision of the dichotomy view, which will be considered in the following section.

Challenges to the preattentive/attentive distinction

Callaghan (1986, 1989; Callaghan, Lasaga & Garner, 1988) explored the limits of preattentive processing using a texture segregation task with adults. Texture segregation requires the observer to make boundary judgements for multielement displays and is widely used as an indicator of preattentive processing (e.g., Beck, 1972, 1982; Julesz, 1975, 1981, Treisman & Gelade, 1980). The logic of Callaghan's experiments is that if preattentive processing is qualitatively distinct from attentive processing (i.e., based on independent as opposed to combined feature information), then when boundaries are defined by single feature differences, all other activity in the texture display (i.e., activity in irrelevant feature maps) should have no impact on the boundary decision. To test this logic two types of arrays were designed. One for which elements differed across a boundary in a single property alone (control), and another for which elements differed in a single property to define a boundary, but also varied on a second, irrelevant property across the entire display [low-similarity (called orthogonal in Callaghan, 1986 and Callaghan et al, 1988)]. The evidence for a clear absence of attentional processing from this classically preattentive task would be the finding that irrelelvant variation of other featural information across the display did not interfere with boundary decisions. Interference would of course imply that attention was at work in texture segregation and a dissolving of the boundary between preattentive/attentive stages was called for.

The evidence clearly shows strong interference of texture segregation that is symmetric when hue and brightness (Callaghan, 1986) and hue and horizontal/vertical or left/right diagonal orientations (Callaghan et al, 1988) are used as stimuli. Interference is asymmetric for hue and geometric form (Beck, 1983; Callaghan, 1989), and for hue and horizontal/left diagonal orientation stimuli (Callaghan et al, 1988). Furthermore interference is always in favour of hue interfering with the processing of the other dimension, but not vice versa. The findings of Enns (1986) corroborate the claim that the processing of feature information in a texture segregation task is not immune from other information contained in the display. In Enns' experiment ease of segregation was determined by the ratio of unique to common elements in the regions. Likewise, the recent data from Treisman and Gormican's (1988) studies of visual search asymmetries suggest that a reformulation of the stage metaphor is in order. Specifically, the visual search results imply that when featural information is shared by targets and distractors, and search is for a single feature, search will be serial in order that problems of discriminability can be overcome.

In their revision of the feature integration theory Treisman and Gormican (1988) abandon the dichotomy view, and see attention as varying along a continuum during perceptual processing. Attention ranges from widely dispersed, as in texture segregation, to sharply focused on a single item, as in visual search for a conjunction of features in a heterogeneous background of elements that share those features. In this revised view the notion of shifting of the "spotlight" of attention, from wide to narrow focus, bears some resemblance to developmental findings in selective attention tasks.

Developmental attention data and the revised view of perceptual processing

There is a vast literature, based on studies that employ attentional tasks, showing that the organization of perceptual space may be very different for young children as opposed to older children and adults (see Kemler, 1983a, 1983b; Shepp,1978, 1983 for reviews of this literature). The results from these studies support what has become known as the integrality-separability hypothesis. Simply stated, the hypothesis argues that young children tend to organize stimuli that are separable, in the sense of Garner (1974), according to overall similarity relations (i.e., the stimulus is a wholistic unit for them), whereas older children and adults tend to organize these stimuli according to dimensional relations (i.e., the stimulus is composed of analyzable parts). There are corollaries to this hypothesis: (1) All observers, young and old alike, tend to organize integral stimuli (Garner, 1974) according to overall similarity relations, and are apparently unable to attend to component dimensions, except under special conditions of training (Foard & Kemler Nelson, 1984; J.D. Smith & Kemler Nelson, 1984). (2) When separable stimuli are made more complex (e.g., by adding more dimensions to the stimulus set) adults will tend to classify them by overall rather than dimensional relations (L.B. Smith, 1981). (3) Kemler Nelson (1983b) stresses that this hypothesis refers to tendencies of observers at these ages, and it is not meant to imply, for example, that young children are never capable of perceiving dimensional relations, nor adults wholistic relations.

How do these developmental findings relate to the adult data from texture segregation studies? The findings of interference in texture segregation (Callaghan, 1986; 1989; Callaghan et al, 1988) call for a continuum rather than a dichotomy view of attention. Treisman and Gormican (1988) argue that attention will be widely dispersed across the visual field for the task of texture segregation. The nature of the task clearly supports this suggestion. In a texture segregation task the observer is instructed to indicate where a boundary lies, and there is no emphasis placed on the nature of the dimensions employed or which dimension may mediate the boundary at any given time. For example, in Callaghan's experiments all types of stimuli (see Figure 1 for sample stimuli), control and low similarity for both component dimensions, are randomly mixed in a block of trials. The observer is clearly not operating in a focused attention mode. Nevertheless, even in a widely dispersed attentional mode, information from different feature maps, and different dimensional groups of feature maps, can intermingle and potentially influence processing. Treisman (Treisman & Gelade, 1980; Treisman & Schmidt, 1982; Treisman & Paterson, 1984) argues that this intermingling can result in illusory conjunctions, and Callaghan (Callaghan, 1986, 1989, Callaghan et al, 1988) shows that it can interfere with boundary perception based on single feature differences.

A consideration of the integrality-separability hypothesis suggests that the attentional demands placed on the child (or adult) in a given task situation will influence the ease with which a given perceptual structure can be accessed (Kemler, 1983b). Presumably, if attentional demands were eliminated in a task, perceived structure should be identical across ages since there would be no grounds for arguing that the process of feature registration undergoes developmental change once the visual system has matured. On the other hand, if attentional demands are very low, or widely dispersed as in the texture segregation task, any difference in perceived structure observed across ages must reflect an attentional mechanism at work. Thus, developmental data from texture segregation studies can help to clarify the nature of early perceptual processing: Is it preattentive or widely dispersed and attentive?

The present studies explore this developmental question and focus on determining the nature of the perceived structure for stimuli using a texture segregation task with children. Based on the findings from attentional studies, which show that young children are more global and less dimensional in their organization than older children and adults, it may be expected that there may be more, rather than less, interference of boundary perception for young children as compared to older children and adults. However, since these are the first studies to explore texture segregation in young children, the nature of the perceived structure is an open question.

The present research

The developmental research reported here is based on the design of the adult studies reported by Callaghan (1989). There are two major classes of studies in the six experiments. In one class (Experiments 1-4), the question of whether lowering the similarity of elements within a region of texture would interfere with texture segregation was asked, and in a second class (Experiments 5-6) it was determined whether there was dominance of one type of property difference over another when observers choose a boundary in an ambiguous, two-boundary display. Within each class, studies differ in the nature of the dimensions that comprise the stimuli. The general method will be outlined, followed by specific results for each of the experiments.

General method: Interference experiments

In the interference experiments (Experiments 1-4) children were presented with 16-element arrays containing one 4-element quadrant that differed from the rest. Fifteen different children at each of the age levels of 5, 7, and 11 years participated in each of the four experiments. The arrays were drawn with Panatone felt markers (see below for approximate Munsell values) on plastic vellum paper and then mounted on cards containing identical line drawings of a clown with a large tummy to contain the stimulus.

The dimensions that varied for Experiments 1 and 2 were hue and/or geometric form, and for Experiments 3 and 4 were hue and/or line orientation. The values of hue varied in Experiment 1 were the Munsell values 10R 4/12 and 2.5RP 4/12, making an easy hue discrimination (7steps in the Munsell system). In Experiment 2 hue values were 10R 4/12 and 2.5R 4/12, making a hard hue discrimination (3 steps). The same forms varied in both Experiments 1 and 2 were the straight and curved novel geometric forms of Figure 1. In Experiment 3 the hue values of Experiment 1 were varied (10R 4/12, 2.5RP 4/12) along with line orientations horizontal and vertical. In Experiment 4 the hue values of Experiment 2 were varied (10R 4/12, 7.5RP 4/12) along with horizontal and vertical line orientation.

There were two major types of arrays in each experiment, single-property (control and low similarity) and two-property (redundant). The five subtypes of arrays that form these major types are illustrated in Figure 1 for the dimensions hue and form, and generalize to hue and line orientation arrays. A difference of hue is symbolized in all figures by a shading difference. Notice that each single property array allows for segregation according to either hue (hue-control, hue-low similarity) or form (form-control, form low similarity) differences. For both control and low similarity arrays a single property difference defines the boundary. These arrays differ in that for control arrays the value of the second property is held constant across the array, whereas for low similarity arrays it is varied randomly across the entire display. Notice also that two-property arrays (redundant) can potentially be segregated on the basis of either, or both, hue and form differences.

The control, low similarity and redundant arrays correspond respectively (see Callaghan, 1986 for details) to the control, orthogonal and correlated arrays that were used in the attentional experiments of Garner and Felfoldy (1970). Performance differences between arrays can illuminate the nature of perceptual structure. Specifically, the pattern of performance that is characteristic of a wholistic structure (i.e., based on overall similiarity relations) is interference for low similarity and gains for redundant, when each are compared to control arrays. The characteristic pattern for dimensional structure (i.e., based on dimensional relations) is no interference or gain; equivalent performance across all arrays. Furthermore, interference in low similarity as compared to control arrays suggests that featural information is not independent in texture segregation and that processing must involve a component of attention, albeit widely dispersed.

Children were presented with four exemplars of each of the five subtypes of arrays on each of two days of testing. Testing was preceded by practice trials on each day. Children were instructed to point, as quickly as they could without making mistakes, to the corner of the stimulus that differed from the rest. The experimenter, who was highly practiced in this procedure, started a stopwatch as soon as she turned over the stimulus card and stopped it when the child's finger touched one of the quadrants of the stimulus. Errors and reaction time to the nearest hundreth of a second were recorded. Trials were considered mistrials if the child's fixation was diverted from the stimulus at any time from the starting to the stopping of the stopwatch, and were run again at the end of the day's block. Each daily session lasted approximately 12 minutes.

Errors were low (averaging 0.03 across experiments) and did not reflect a speed/accuracy tradeoff, thus were not included in analysis. The data for these experiments are the median response time (RT), calculated for each array subtype, on each day, for each observer's correct responses. In the analysis of these data the following steps are taken. First, it is determined whether the ease of segregation based on the particular values chosen is equivalent across dimensions (i.e., hue-control vs. form-control, or hue-control vs. orientation-control). Then, separate analyses for each component dimension determine whether lowering the similiarity of elements in low similarity arrays interferes with segregation when compared to the appropriate control array (e.g., hue-control vs. hue-low similarity). The pattern of interference that is found is considered in light of the relative discriminabilities of control arrays. Finally, the question of whether there is a redundancy gain in performance is assessed by comparing each individual's RT for redundant arrays with the faster RT of their control arrays. The pattern of performance across all these arrays will then clarify whether the percieved structure is wholistic (i.e., interference for low similarity arrays and gain for redundant arrays) or based on dimensions (i.e., equivalent performance across all arrays).

Results: Interference experiments

Experiment 1: Easy hue and geometric form

The analysis for control arrays revealed main effects of age, days, and conditions. Thus, there was an increase in RT (sec) that was correlated with age (5 years = 1.70, 7 years = 1.07, 11 years = 0.88), a decrease in RT from Day 1 to Day 2 (Day 1 = 1.28, Day 2 = 1.15), and most importantly, evidence that hue-control arrays were easier to segregate than form-control arrays (hue-control = 1.14, form-control = 1.29). There were no significant interactions.

In the analysis for interference of hue arrays there were significant main effects of age and days, and an interaction of age *x* condition. The day effect showed that RT was faster on Day 2 (1.11) as compared to Day 1 (1.23), and the age effect is qualified by the interaction. A posteriori tests of the interaction revealed that form variation interfered with segregation based on hue only for 5-year olds (hue-control = 1.62, hue-low similarity = 1.79).

In the analysis of interference for form arrays there were main effects of age, days, and conditions, and interactions of age *x* condition, and day *x* condition. A posteriori tests of the interactions showed that there was interference at all ages (hue variation interfering with form segregation), but that the effect was larger for 5- and 7-year olds than 11-year olds. In addition, the interference effect was significant on both days, but was larger on Day 1 (form-control = 1.36, form-low similarity = 1.94) than on Day 2 (form-control = 1.22, form-low similarity = 1.52).

In the analysis of redundant arrays there were expected main effects of age, and days. There was no evidence of a redundancy gain. Failure to find redundancy gains when there are interference effects is not unusual, especially when the stimuli are easily discriminated (Callaghan, 1989; Garner, 1983). Thus, in the interpretation of the relevance of these patterns for perceptual structure we may argue for wholistic structure, even when there are no redundancy gains.

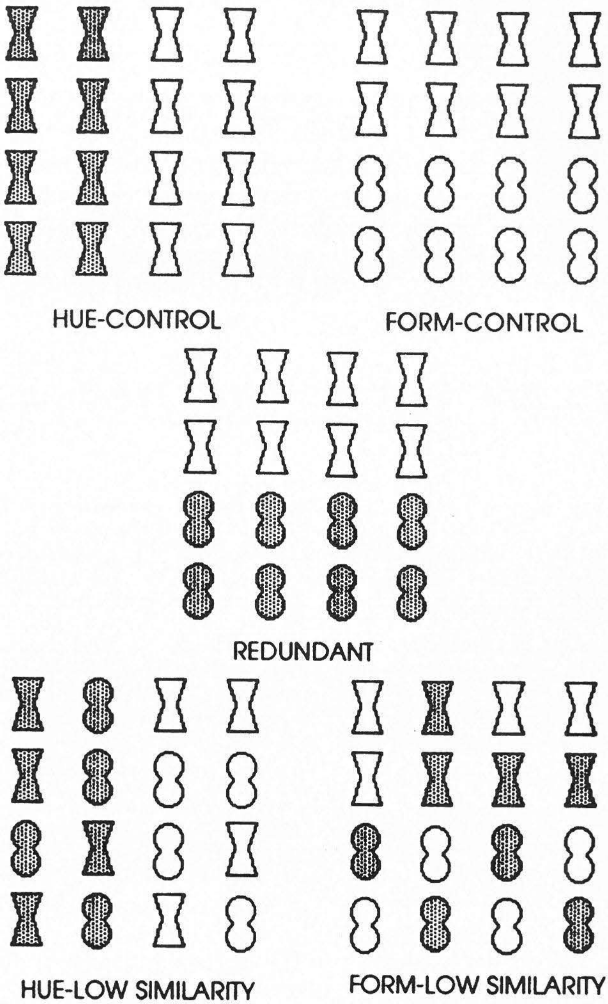

Figure 1. Subtypes of stimulus arrays used in Experiments 1 and 2.

With these stimuli we find that 5-year olds show symmetric interference (i.e., hue interferes with form and vice versa), 7-year olds and 11-year olds show asymmetric interference with hue interfering with form judgments but form not interfering with hue judgments. This pattern of asymmetry is also characteristic of adults (Callaghan, 1989). Thus, if we relax the criterion of redundancy gain, perceptual structure for 5-year olds is more glued-together, or wholistic than it is for 7- and 11-year olds. The structure for the latter two age groups seems to fall between wholistic and dimensional; it is not strictly either. The reason for hue interference with the present stimuli could be that hue was easier to discriminate than form; this hypothesis was tested in Experiment 2, where hue was made harder to discriminate.

Experiment 2: Hard hue and geometric form

The analysis for control arrays revealed main effects of age, days, and conditions, and an interaction of days *x* conditions. There was an increase in RT that was correlated with age (5 year = 1.63, 7 years = 1.46, 11 years = 1.05), a decrease in RT from Day 1 to Day 2 (Day 1 = 1.47, Day 2 = 1.29), and again evidence that hue-control arrays were easier to segregate than form-control arrays (hue-control = 1.35, form-control = 1.42). The differential in discriminability between hue and form arrays has been reduced considerably compared to Experiment 1 (difference of 0.06 Experiment 2 and 0.15 Experiment 1). Here it is a small difference, though a significant one. The interaction revealed that the improvement in performance for form-control arrays from Day 1 to Day 2 (1.53 vs. 1.30) was larger than the improvement (1.41 vs. 1.29) for hue-control arrays. Thus, although we increased the difficulty of the hue discrimination it did not reach equivalence with form discriminability as we had hoped.

In the analysis for interference of hue arrays there were significant main effects of age, days, and conditions. There was improvement over age (5 years = 1.61, 7 years = 1.46, 11 years = 1.07). The day effect showed that RT was faster on Day 2 (1.31) as compared to Day 1 (1.45), and the condition effect revealed significant interference from form variation at all ages (hue-control = 1.35, hue-low similarity = 1.41). In the analysis of interference for form arrays there were main effects of age, days, and conditions, and an interaction of day *x* condition. The interaction is showing that there was improvement in performance for form-control arrays from Day 1 to Day 2 (1.53 vs. 1.30) but not for form-low similarity arrays (2.00 vs. 2.05). Interference from hue variation was strong on both days.

In the analysis of redundant arrays there were expected main effects of age and days, and an interaction of age *x* condition. Inspection of the interaction revealed that there was a redundancy gain only for the 5-year olds.

With these stimuli, for which hue is relatively hard to discriminate we find symmetric interference (i.e., hue interferes with form and vice versa) at all ages. This leads us to the conclusion that the preferred organization for children with these stimuli in a texture segregation task is wholistic, however firm indication of this must await data from arrays for which hue is equivalent to, or harder than, form.

Experiment 3: Easy hue and line orientation

The analysis for control arrays revealed main effects of age, days, and conditions. There was an increase in RT that was correlated with age (5 years = 1.53, 7 years = 1.04, 11 years = 0.83), a decrease in RT from Day 1 to Day 2 (Day 1 = 1.17, Day 2 = 1.10), and evidence that hue-control arrays were easier to segregate than orientation-control arrays (hue-control = 1.10, form-control = 1.17). The differential in discriminability between hue and orientation values is small, though significant.

In the analysis for interference of hue arrays there were significant main effects of age and days. There was improvement over age (5 years = 1.55, 7 years = 0.97, 11 years = 0.83) and days (Day 1 = 1.16, Day 2 = 1.08). In the analysis of interference for orientation arrays there were main effects of age, days, and conditions. There was improvement over age (5 years = 1.83, 7 years = 1.32, 11 years = 0.94), and days (Day 1 = 1.41, Day 2 = 1.31). Analysis of form arrays indicated that interference from hue variation was strong (orientation-control = 1.17, orientation-low similarity = 1.55). The only significant effect in the analysis of redundant arrays was a main effect of age (5 years = 1.42, 7 years = 0.93, 11 years = 0.79).

With these stimuli we find asymmetric interference (i.e., hue variation interferes with orientation segregation, but orientation does not interfere with hue) at all ages. As in Experiment 1, hue was the more discriminable of the two dimensions, thus although the data more readily suggest a wholistic than dimensional structure, we hold off firm conclusions until considering the data of the next experiment, where hue discriminability was harder than orientation.

Experiment 4: Hard hue and line orientation

The analysis for control arrays revealed main effects of age, days, and conditions. There was an increase in RT that was correlated with age (5 years = 1.85, 7 years = 1.51, 11 years = 1.01), a decrease in RT from Day 1 to Day 2 (Day 1 = 1.57, Day 2 = 1.35), and evidence that hue-control arrays were harder to segregate than orientation-control arrays (hue-control = 1.52, orientation-control = 1.40). Thus, the differential in discriminability between hue and orientation arrays has been reversed as compared to Experiment 3. We now consider the patterns of interference with this in mind.

In the analysis for interference of hue arrays there were significant main effects of age, days, and conditions, and an interaction of age *x* conditions. There was improvement over age (5 years = 2.13, 7 years = 1.73, 11 years = 1.12) and days (Day 1 = 1.78, Day 2 = 1.54). Analysis of the interaction revealed that there was significant interference of hue segregation for 5- and 7-year olds, but not for 11-year olds.

In the analysis of interference for orientation arrays there were main effects of age, days, and conditions, as well as interactions of age *x* conditions, and conditions *x* days. Analyses of interactions showed that interference was stronger on Day 1 (orientation-control = 1.51, orientation-low similarity = 1.91) than Day 2 (orientation-control = 1.28, orientation-low similarity = 1.49), and that there was stronger interference for 5- and 7-year olds as compared to 11-year olds.

In the analysis of redundant arrays, there were significant main effects of age (5 years = 1.42, 7 years = 0.93, 11 years = 0.79), and days (Day 1 = 1.39, Day 2 = 1.25), and no evidence of redundancy gains in performance.

With these stimuli, for which hue is hard to discriminate relative to orientation, we find symmetric interference (i.e., hue interferes with orientation and vice versa) for 5- and 7-year olds, and asymmetric interference for 11-year olds. Thus, even though hue is harder to discriminate than orientation it still interferes with segregation based on the latter dimension. This findings strongly suggest that the preferred organization for 5- and 7-year olds with these stimuli in a texture segregation task is wholistic, and more weakly (since interference was asymmetric) suggest that the preferred organization for 11-year olds is, as in Experiment 2, more wholistic than dimensional.

Conclusions for Experiments 1-4

Taken together the findings of the first four experiments suggest that the wholistic structure of the stimulus predominates in texture segregation tasks. With all stimuli there was some degree of interference at all ages. In the attentional literature (Garner, 1974, Garner & Felfoldy, 1970) interference, or failure of selective attention, has been taken as indication that the dimensions of the stimulus are 'glued together' for the observers. Typically, the criterion of redundancy gain has also been required to show wholistic structure (Garner, 1974), however, both its' necessity and elusiveness have recently been questioned (Garner, 1983). In our experiments it has also proven to be elusive.

A question of intrigue that is raised by these results is why there should be asymmetric interference, and why the asymmetry should always favour hue, as it did for adults in Callaghan's (1989) experiments? One possibility raised by Callaghan was that the dimension of hue may dominate, or have a priviledged status, early in perception. The next two experiments test this claim for children.

General method: Dominance experiments

The stimuli for these experiments were constructed from the same materials, and used the same dimensions, as the previous experiments. Stimuli were again 16-element arrays that contained either one (control) or two (ambiguous) quadrants that differed from the rest. Ten different children at each of the age levels 5, 7, and 11 years participated in each of the two dominance experiments.

In Experiment 5 stimuli were formed from hue and/or geometric form (straight vs. curved novel forms of Experiments 1 and 2) dimensions, and in Experiment 6 from hue and/or line orientation (horizontal vs. vertical). Hue had three possible discriminability levels in experiments 5 and 6: Hard (10R 4/12 vs. 2.5R 4/12, 3 Munsell steps), Medium (10R 4/12 vs. 7.5RP 4/12, 5 Munsell steps) and Easy (10R 4/12 vs. 2.5RP 4/12, 7 Munsell steps).

There were two possible array types in Experiments 5 and 6, single-boundary (control) and two-boundary (ambiguous). These are illustrated in Figure 2 for hue and line orientation stimuli, and generalize to hue and form arrays. Notice that single-boundary arrays could have boundaries defined either by hue (hue-control) or (Experiment 5) form (form-control) differences, or hue or (Experiment 6) line orientation (orientation-control) differences. When hue differences defined the boundary, these differences could be either Hard, Medium, or Easy to discriminate, according to the hue values listed above. Thus, there were three subtypes of hue-control arrays. There was only one type of form (straight vs. curved novel form) or line orientation (horizontal vs. vertical) control array. Notice also that two-boundary arrays have two potential regions that differ from the background, one region is defined by hue differences and the other by orientation (or form) differences. There were three subtypes of ambiguous arrays, corresponding to the discriminability of the elements that produced the hue boundary. These three levels were identical to the Hard, Medium, and Easy levels of discriminability in the hue-control arrays. The orientation differences, or form differences, that produced the second boundary in ambiguous arrays was the same for all subtypes of ambiguous arrays. By holding one dimensional difference at a constant level of discriminability (i.e., orientation or form) and varying the discriminability of the other (i.e., hue) we hoped to track whether one or the other dimension dominated in the child's boundary decision.

Children were presented with four exemplars of each type of hue-control (Hard, Medium, Easy) and ambiguous (Hard, Medium, Easy) arrays on each of three days of testing. In order to ensure that an imbalance of hue-control to form (or orientation)-

control arrays did not produce a bias in favour of choosing hue boundaries in ambiguous arrays, additional form(orientation)-control arrays were included in the deck such that there were equal numbers (total of 12) of hue-control and form-control (or orientation-control) arrays presented in each daily session. Children were instructed to point, as quickly as possible without mistakes, to the corner of the stimulus that differed from the rest. In the case of ambiguous arrays, they were told to choose the one corner that stood out the most as being different. As in the previous experiments, the experimenter recorded errors and RT (hundreths of a sec), and ran mistrials at the end of each session. Each daily session lasted approximately 15 minutes.

The following steps were taken in the analysis of the data for the dominance experiments. First, it was determined when (i.e., Hard, Medium, or Easy hue-control) discriminability of form(or orientation)-control arrays equalled that of hue. Median RTs for correct responses in each subtype of control array (hard hue-control, medium hue-control, easy hue-control, form/orientation-control), collapsed across days, served as data for an analysis of variance. Then, the question of dominance was assessed by looking at the average proportion of choice for hue as compared to that for form (or orientation) in ambiguous arrays. Separate analyses of variance were then conducted for each dimension. The pattern of performance that indicates dominance would be overwhelming choice for one dimension over the other that holds regardless of relative discriminability between dimensions. In contrast, equal status of dimensions would be revealed by a pattern of choice that was guided by relative discriminability of the component dimensions. Thus, choice for hue when it was more discriminable, and for form or orientation when they were more discriminable.

Results: Dominance experiments

Experiment 5: Hue and form ambiguous

The data relevant to these analyses are graphed in Figure 3. Note that each row of this figure contains data for a given age group, and that at each age level the RT data for control arrays are graphed separately from the data showing proportion of choice in ambiguous arrays. In the analysis of control data there were significant age and condition main effects. The trends for age show an unexpected superiority of 7-year olds over 11-year olds. We do not have an explanation for this finding. The condition effects indicate that form-control is equivalent to hard hue-control for all ages. With this in mind, let us consider the choice data.

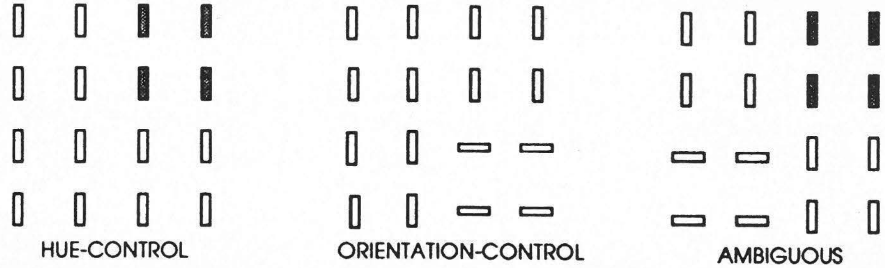

Figure 2. **Subtypes of stimulus arrays used in Dominance Experiments 6.**

When form is equal to hue in discriminability for these stimuli (i.e., hard hue) analyses indicated that choice for form and hue were at chance level for 5- and 11-year olds, and slightly above chance for 7-year olds. This pattern indicates no dominance of one dimension over the other when they are of equivalent discriminability. When hue is easier to discriminate (i.e., medium and easy hue) analysis showed that choice for hue boundaries was above chance, and for form boundaries was below chance for children at all ages. In sum, the data strongly suggest that there is no dominance of hue over form in texture segregation tasks, but that children choose boundaries based on the most discriminable dimension. These results mirror the findings with adults (Callaghan, 1989).

Experiment 6: Hue and line orientation ambiguous

The data relevant to these stimuli are graphed in Figure 4. In the analysis of control data significant age and condition main effects are qualified by an age *x* condition interaction. Analysis of the interaction reveals that for 5-year olds orientation-control arrays are equivalent to hard hue-control arrays, and both are more difficult to discriminate than medium and easy hue-control arrays. For 7-year olds orientation-control arrays are equivalent to medium hue-control arrays, are easier than hard hue-control, and harder than easy hue-control arrays. For 11-year olds orientation-control arrays did not differ significantly from any of the hue-control arrays, however hard hue-control was more difficult to discriminate than easy hue-control. With these trends in mind, let us consider the choice data.

First consider the 5-year olds. When orientation is equal to hue in discriminability for these stimuli (i.e., hard hue) analyses indicated that choice for hue is significantly above chance and increases as hue becomes more discriminable. Choice for orientation is well below chance when hue and orientation are equal in discriminability, and decreases as hue becomes more discriminable. This pattern looks like dominance for hue over orientation when boundaries based on both are present in the array.

Next consider the 7-year olds. For these children orientation discriminability equals that of hue when hue has a medium discriminability level. At this point choice for orientation and for hue boundaries are exactly at chance level. When hue levels are harder to discriminate (hard hue), then choice for hue is below chance and for orientation above chance. When hue levels are easier than orientation to discriminate (easy hue), then choice for hue is above chance, and for orientation below chance. This is precisely the pattern that indicates no dominance of one dimension over the other. Choice is guided by the most discriminable dimension.

Finally, consider the 11-year olds. Unfortunately since we cannot discriminate between the control conditions for these children a clear indication of the status of dimensions cannot be obtained. However, looking at the pattern for choice shows that as hue becomes physically more discriminable (even the 11-year olds show a trend from less to more discriminable as you go from hard to easy hue), the the proportion of choice for that dimension goes from chance (hard and medium hue) to well above chance (easy hue).

In sum, there appears to be age differences when the dominance question is considered for hue and orientation stimuli. The pattern for 5-year olds suggests that hue dominates orientation in texture segregation judgments. In contrast, the pattern for 7-year olds, and as far as we can tell for 11-year olds, indicates that there is no dominance. Choice is based on the most discriminable dimension.

Conclusions for the dominance experiments

With the exception of 5-year olds confronted with hue and orientation stimuli, there appears to be no dominance of dimensions in texture segregation. This is consistent with adult findings (Callaghan, 1989), and with the generally accepted view by researchers in the area. It is usually believed that basic visual features (visual primitives in Treisman's terms) mediate texture segregation, and that no one of these features is more basic, or primitive than the others. The only result in the literature that may bear on the apparent dominance of hue over orientation for 5-year olds is the finding by von Wright (1968) that in a partial report visual iconic memory task one can easily select by hue but not by orientation. This suggests that early in processing hue may be more accessible than orientation, however why we find evidence for this only for 5-year olds in the texture segregation task is not clear.

General discussion

The impetus for these studies comes from recent research with adults suggesting that the dichotomy of preattentive/attentive may be no longer useful when applied to stages of visual processing (Beck, 1983; Callaghan, 1986, 1989, Callaghan et al , 1988; Treisman & Gormican, 1988). It was hoped that data from developmental studies of texture segregation would help to clarify what is becoming evident from the adult studies; that is, texture segregation involves some degree of attentional processing. It was predicted that there would be interference of segregation when values of an irrelevant dimension varied across the stimulus, that the observer would not be able to 'shut out' this information.

The results support this prediction. With the exception of 11-year olds in Experiment 4, children of all ages showed symmetric interference for both types of stimuli (i.e., hue and form, and hue and line orientation). Thus, they have difficulty ignoring irrelevant information when making a segregation response. The finding of symmetric interference for hue and form stimuli at all ages in Experiment 2 contrasts with the adult trend of asymmetric interference reported by Callaghan (1989). Thus, the degree of interference is greater for children than adults with these stimuli. This would be expected if attentional mechanisms were invoked during the segregation response.

Taken together, the results of the present research strengthen the claim that although texture segregation may occur relatively early in perception, it is not a pre-attentive task. Although there was little indication of dimensional dominance in these data (only the 5-year olds with hue and orientation stimuli), future studies that employ a variety of tasks may help to clarify whether hue does in fact have a priviledged status when compared to orientation early in perception.

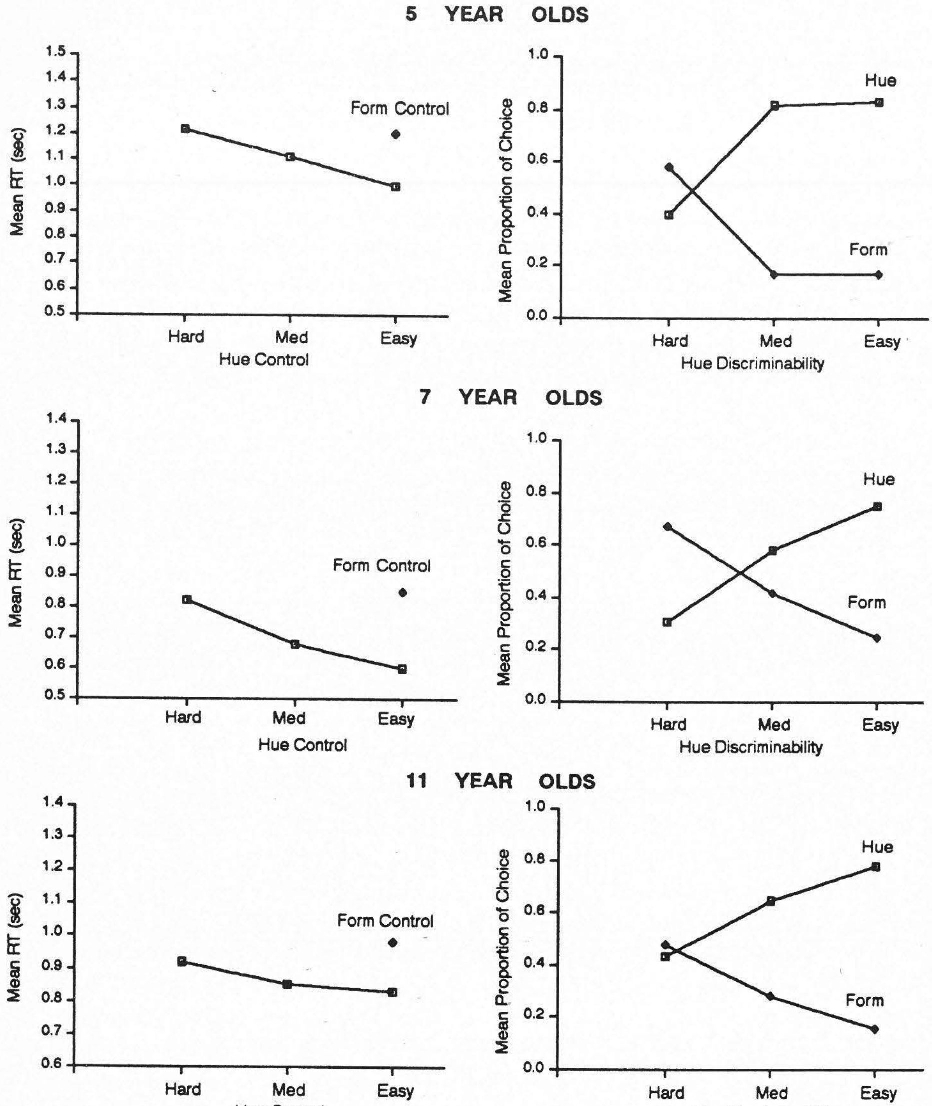

Figure 3. **Developmental data for the Hue/Form Ambiguous Experiment 5.**

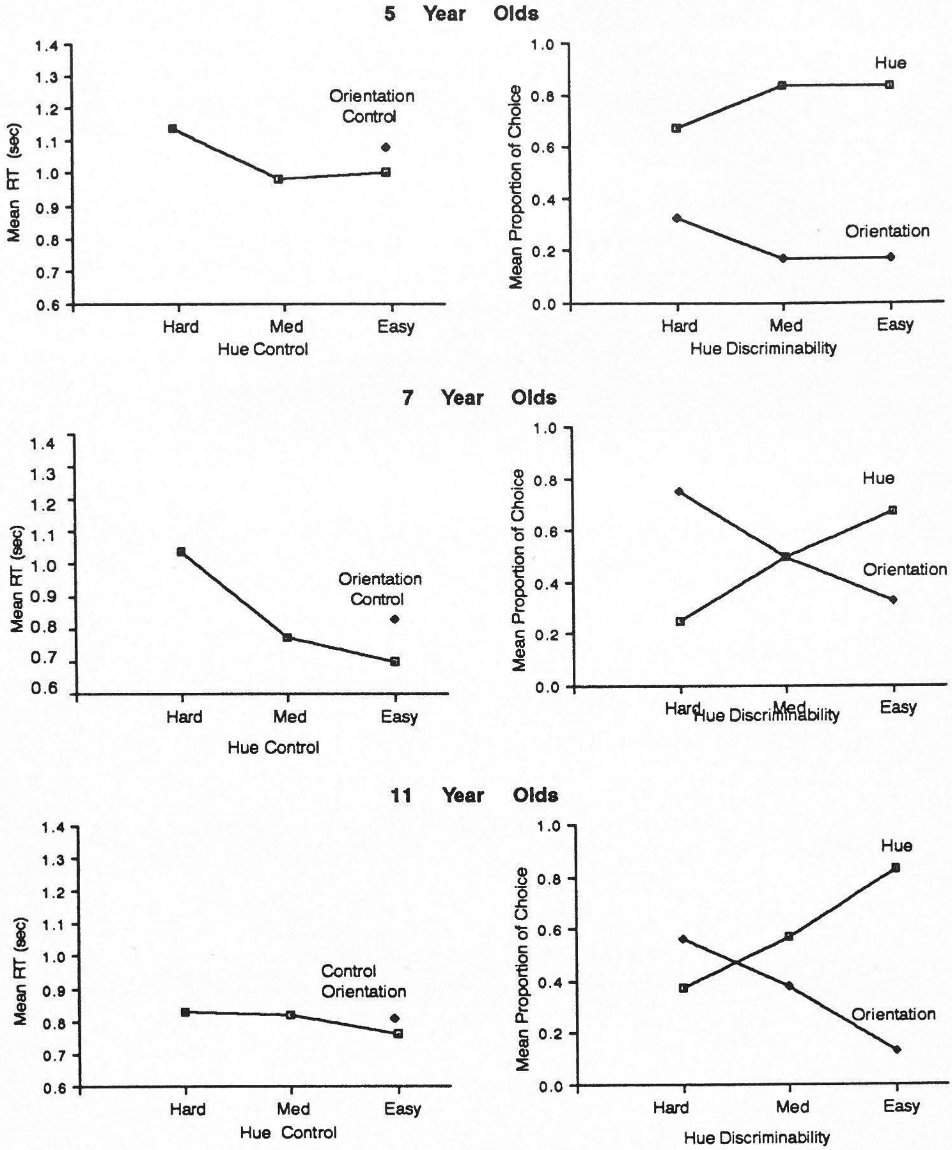

Figure 4. **Developmental data for the Hue/Orientation Ambiguous Experiment 6.**

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